# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679


## ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 50 | $d = 3$, $a = 5$; $t_{16} = a + 15d$. |
| 2 | **(a)** 37 | $t_k = S_k - S_{k-1} = p(2k-1) + q$. |
| 3 | **(c)** 816 | AP with $a = 6$, $d = 6$. |
| 4 | **(a)** 7 | $\frac{n}{2}[2a + (n-1)d] = 510$ gives $d$. |
| 5 | **(b)** 14 | Take $m - d,\ m,\ m + d$; the sum gives the middle number. |
| 6 | **(a)** 12100 | Yearly values form a GP with ratio $1 + \frac{r}{100}$. |
| 7 | **(b)** 2 | $r^7 = 128$. |
| 8 | **(d)** 440 | Add the first and last term, multiply by $\frac{n}{2}$. |
| 9 | **(a)** $-4$ | $d = -4$, $a = 60$; $t_{17} = a + 16d$. |
| 10 | **(d)** 70 | $t_k = S_k - S_{k-1} = p(2k-1) + q$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\frac{n}{2}[2a + (n-1)d] = 255$; $d = 2$ (1); $t_{15} = 31$ (1). | 1+1 |
| 12 | $a + (n-1)d > 200$; $(n-1) > 32.50$ (1); $n = 34$ (1). | 1+1 |
| 13 | $t_1 = S_1 = 7$ (1); $t_{10} = S_{10} - S_9 = 61$ (1). | 1+1 |
| 14 | $\frac{n}{2}[2a + (n-1)d] = 336$; $d = -4$ (1); $t_{12} = 6$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $2a + 10d = 40$ and $a + 4d = 17$ (1); Solving: $a = 5$, $d = 3$ (1); $S_{20} = 670$ (1). | 1+1+1 |
| 16 | Terms: $12, 17, 22, \ldots, 197$ (1); Number of terms $= 38$ (1); Sum $= \frac{n}{2}(a + l) = 3971$ (1). | 1+1+1 |
| 17 | Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 216$, so $m = 6$ (1); $\frac{6}{r} + 6r = 15$ gives $r = 2$ (1); Numbers 3, 6, 12 (1). | 1+1+1 |
| | **OR:** $2a + 10d = 64$ and $a + 4d = 28$ (1); Solving: $a = 12$, $d = 4$ (1); $S_{14} = 532$ (1). | 1+1+1 |
| 18 | Terms: $24, 31, 38, \ldots, 297$ (1); Number of terms $= 40$ (1); Sum $= \frac{n}{2}(a + l) = 6420$ (1). | 1+1+1 |
| | **OR:** Let them be $\frac{m}{r},\ m,\ mr$; Product $= m^3 = 1728$, so $m = 12$ (1); $\frac{12}{r} + 12r = 40$ gives $r = 3$ (1); Numbers 4, 12, 36 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Penalties form an AP (1); $t_{30} = 200 + 29 \times 50 = 1650$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{30} = \frac{30}{2}(200 + 1650)$ (1); Total ₹27750 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{2}{3}$ is constant, so a GP (1); 1st rebound $= 54$ m (1); 3rd rebound $= h \times (2/3)^3$ (1); $= 24$ m (2). | 1+1+1+2 |
| 20 | Penalties form an AP (1); $t_{24} = 100 + 23 \times 25 = 675$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{24} = \frac{24}{2}(100 + 675)$ (1); Total ₹9300 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{3}{4}$ is constant, so a GP (1); 1st rebound $= 48$ m (1); 3rd rebound $= h \times (3/4)^3$ (1); $= 27$ m (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(b)** 2812 | AP with $a = 4$, $d = 4$. |
| 2 | **(c)** $-3$ | $\frac{n}{2}[2a + (n-1)d] = 147$ gives $d$. |
| 3 | **(a)** 17 | Take $m - d$, $m$, $m + d$; the sum gives the middle number. |
| 4 | **(a)** 28800 | Yearly values form a GP with ratio $1 + \frac{r}{100}$. |
| 5 | **(d)** 3 | $r^5 = 243$. |
| 6 | **(b)** 319 | Add the first and last term, multiply by $\frac{n}{2}$. |
| 7 | **(d)** 97 | $d = 9$, $a = 7$; $t_{11} = a + 10d$. |
| 8 | **(d)** 57 | $t_k = S_k - S_{k-1} = p(2k-1) + q$. |
| 9 | **(d)** 2277 | AP with $a = 9$, $d = 9$. |
| 10 | **(c)** 5 | $\frac{n}{2}[2a + (n-1)d] = 927$ gives $d$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $a + (n-1)d > 150$; $(n-1) > 19.86$ (1); $n = 21$ (1). | 1+1 |
| 12 | $t_1 = S_1 = 7$ (1); $t_{12} = S_{12} - S_{11} = 51$ (1). | 1+1 |
| 13 | $\frac{n}{2}[2a + (n-1)d] = 1218$; $d = 5$ (1); $t_{21} = 108$ (1). | 1+1 |
| 14 | $a + (n-1)d > 301$; $(n-1) > 33.22$ (1); $n = 35$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $2a + 10d = 62$ and $a + 6d = 38$ (1); Solving: $a = -4$, $d = 7$ (1); $S_{12} = 414$ (1). | 1+1+1 |
| 16 | Terms: $17, 21, 25, \ldots, 149$ (1); Number of terms $= 34$ (1); Sum $= \frac{n}{2}(a + l) = 2822$ (1). | 1+1+1 |
| 17 | Let them be $\frac{m}{r}$, $m$, $mr$; Product $= m^3 = 8000$, so $m = 20$ (1); $\frac{20}{r} + 20r = 85$ gives $r = 4$ (1); Numbers 5, 20, 80 (1). | 1+1+1 |
| | **OR:** $2a + 14d = 32$ and $a + 8d = 14$ (1); Solving: $a = 30$, $d = -2$ (1); $S_{16} = 240$ (1). | 1+1+1 |
| 18 | Terms: $31, 40, 49, \ldots, 247$ (1); Number of terms $= 25$ (1); Sum $= \frac{n}{2}(a + l) = 3475$ (1). | 1+1+1 |
| | **OR:** Let them be $\frac{m}{r}$, $m$, $mr$; Product $= m^3 = 3375$, so $m = 15$ (1); $\frac{15}{r} + 15r = 50$ gives $r = 3$ (1); Numbers 5, 15, 45 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Penalties form an AP (1); $t_{18} = 500 + 17 \times 100 = 2200$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{18} = \frac{18}{2}(500 + 2200)$ (1); Total ₹24300 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{4}{5}$ is constant, so a GP (1); 1st rebound $= 100$ m (1); 3rd rebound $= h \times (4/5)^3$ (1); $= 64$ m (2). | 1+1+1+2 |
| 20 | Penalties form an AP (1); $t_{28} = 150 + 27 \times 30 = 960$ (1); $S_n = \frac{n}{2}(a + l)$ (1); $S_{28} = \frac{28}{2}(150 + 960)$ (1); Total ₹15540 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{3}{4}$ is constant, so a GP (1); 1st rebound $= 192$ m (1); 3rd rebound $= h \times (3/4)^3$ (1); $= 108$ m (2). | 1+1+1+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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## ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sequences and Progressions** | Class 9 Maths | Maximum Marks: 40

### SECTION — A (Objective Type) — Answers — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 22 | Take $m-d,\ m,\ m+d$; the sum gives the middle number. |
| 2 | **(c)** 6655 | Yearly values form a GP with ratio $1+\frac{r}{100}$. |
| 3 | **(a)** 2 | $r^8 = 256$. |
| 4 | **(a)** 560 | Add the first and last term, multiply by $\frac{n}{2}$. |
| 5 | **(a)** 57 | $d=6,\ a=-15;\ t_{13}=a+12d$. |
| 6 | **(a)** 35 | $t_k = S_k - S_{k-1} = p(2k-1)+q$. |
| 7 | **(d)** 3588 | AP with $a=13,\ d=13$. |
| 8 | **(d)** 4 | $\frac{n}{2}[2a+(n-1)d]=640$ gives $d$. |
| 9 | **(b)** 12 | Take $m-d,\ m,\ m+d$; the sum gives the middle number. |
| 10 | **(d)** 44100 | Yearly values form a GP with ratio $1+\frac{r}{100}$. |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $t_1 = S_1 = 6$ (1); $t_8 = S_8 - S_7 = 76$ (1). | 1+1 |
| 12 | $\frac{n}{2}[2a+(n-1)d]=297$; $d=3$ (1); $t_{18}=42$ (1). | 1+1 |
| 13 | $a+(n-1)d>111$; $(n-1)>24.25$ (1); $n=26$ (1). | 1+1 |
| 14 | $t_1 = S_1 = 11$ (1); $t_{15} = S_{15} - S_{14} = 123$ (1). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $2a+8d=64$ and $a+9d=62$ (1); Solving: $a=8,\ d=6$ (1); $S_{13}=572$ (1). | 1+1+1 |
| 16 | Terms: $11, 17, 23, \ldots, 179$ (1); Number of terms $=29$ (1); Sum $=\frac{n}{2}(a+l)=2755$ (1). | 1+1+1 |
| 17 | Let them be $\frac{m}{r},\ m,\ mr$; Product $=m^3=13824$, so $m=24$ (1); $\frac{24}{r}+24r=60$ gives $r=2$ (1); Numbers 12, 24, 48 (1). | 1+1+1 |
| | **OR:** $2a+16d=84$ and $a+3d=17$ (1); Solving: $a=2,\ d=5$ (1); $S_{25}=1550$ (1). | 1+1+1 |
| 18 | Terms: $43, 51, 59, \ldots, 259$ (1); Number of terms $=28$ (1); Sum $=\frac{n}{2}(a+l)=4228$ (1). | 1+1+1 |
| | **OR:** Let them be $\frac{m}{r},\ m,\ mr$; Product $=m^3=27000$, so $m=30$ (1); $\frac{30}{r}+30r=156$ gives $r=5$ (1); Numbers 6, 30, 150 (1). | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Penalties form an AP (1); $t_{35}=80+34\times 20=760$ (1); $S_n=\frac{n}{2}(a+l)$ (1); $S_{35}=\frac{35}{2}(80+760)$ (1); Total ₹14700 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{4}{5}$ is constant, so a GP (1); 1st rebound $=800$ m (1); 3rd rebound $=h\times(4/5)^3$ (1); $=512$ m (2). | 1+1+1+2 |
| 20 | Penalties form an AP (1); $t_{22}=250+21\times 40=1090$ (1); $S_n=\frac{n}{2}(a+l)$ (1); $S_{22}=\frac{22}{2}(250+1090)$ (1); Total ₹14740 (1). | 1+1+1+1+1 |
| | **OR:** Ratio $\frac{2}{3}$ is constant, so a GP (1); 1st rebound $=108$ m (1); 3rd rebound $=h\times(2/3)^3$ (1); $=48$ m (2). | 1+1+1+2 |

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